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Errors Are The Stepping Stones to Learning: Trial-by-Trial Modeling Reveals Overwhelming Evidence for Mediator Retrievals of Previous Errors in Memory Consolidation

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Abstract

Recent studies suggest that errors facilitate learning in certain conditions. Despite this, reinforcement paradigms dominate learning methods, subscribing to the narrative that errorless learning is the foundation of an ideal learning environment. If we continue to view learning from this restrictive perspective, we may fail to capture and apply the benefits of errors. In this paper, we investigate two potential mechanisms of learning from errors. Participants ($N = 61$) learned word pairs in either a study or error trial before taking a final test. Supporting past error learning literature, errors before a study opportunity led to better performance on a final test. Differences in reaction times between conditions support the theory that errors increase learning by acting as a mediator, or secondary cue, to the correct answer on subsequent tests. Individual differences in model fit using log-likelihood trial-by-trial calculations solidified support for the mediator method.

Keywords: learning from errors; memory; retrieval; elaboration; mediation; computational models; ACT-R

Errors Are The Stepping Stones to Learning

Many people believe that ideal learning is errorless. Errors are often viewed as detrimental to learning with the expectation that they will interfere with the future retrieval of correct information (Ceraso, 1967). These concepts stem from studying more procedural behavior where learning is a constant reinforcement process; thus, errorless learning minimizes opportunities to reinforce errors. This understanding has biased how we optimize learning, backing a paradigm in which learners re-study concepts before even being given a chance to commit a memory error. However, modern-day experimental findings suggest that errors do not pollute the learning process.

Posing a question before revealing the answer, also known as retrieval practice, appears to enable stronger learning. Different experimental manipulations all seem to share a common thread: although the difficulty of retrieval practice impairs performance in the very first few trials, it ultimately results in better final test performance. This has been shown in multiple-choice versus short-answer questions (Greving & Richter, 2018), problems based on a single lesson or spaced across multiple (Rohrer & Taylor, 2007), and retrieval schedules (wide spacing of retrieval attempts vs. short spacing; Roediger & Karpicke, 2010). This widespread finding suggests that errors made early in the learning process from retrieval difficulty can result in better learning.

Additionally, reinforcement paradigms would predict that errors made with higher confidence would reduce learning. Such phenomena would be consistent with interference theories of memory, positing that incorrect items with more substantial traces are likely to produce more interference when attempting to retrieve a correct item (Anderson & Reder, 1999). However, experimental results show that errors committed with high confidence are more likely

to be successfully corrected (Butterfield & Metcalfe, 2001). This suggests that more processing between the onset of a question and the presentation of the answer (i.e., a retrieval attempt) leads to better answer processing, regardless of whether the attempt was successful (Kornell & Vaughn, 2016). Such findings may demonstrate that the behaviorist approach does not translate well to learning by memorization compared to more procedural learning. Achieving successful memorization requires the ability to retrieve information; errors may enhance this type of learning by promoting stronger encoding and thus, more successful recall. In this way, we may be able to enhance the most common form of learning for humans by critically examining how and when errors are beneficial to memorization and subsequent retrieval.

Researchers have introduced new paradigms to study post-error learning, or improvements in recall on subsequent fact presentations after an incorrect answer or memory errors, such as the pretesting paradigm. In one of these paradigms, participants generate an answer before studying it; with no previous exposure to the correct answer, participants are highly likely to generate an incorrect response (Mera, Rodriguez, & Marin-Garcia, 2021). Although this paradigm encourages errors, pretesting for information is still more beneficial than simply studying it. In a paired-associate task, retrieving a free associate of the cue word before learning the correct targets produces better results on a final test than studying the cue and target alone (Kornell, Hays, & Bjork, 2009). Many studies have confirmed this finding, extending the benefits of pretesting to real-world materials (trivia questions; Kornell, 2014), educational materials (math problems; Kapur & Bielaczyc, 2012), and older adults (Cyr & Anderson, 2015).

Although this phenomenon, known as post-error learning, is now well-documented, theinvestigation into its underlying mechanisms is sparse. This line of research has given rise to two prominent theories: the elaborative hypothesis and the mediator hypothesis. To go beyond

speculation, we must empirically examine both of them to successfully leverage post-error learning.

The Elaborative Hypothesis

The elaborative theory of post-error learning posits that unsuccessful retrieval attempts allow for a richer encoding of the correct answer. Also known as the Search Set theory, it poses that retrieval attempts (regardless of success) activate a variety of semantically related candidates. The search set contains many candidates, one of which is the correct answer, thus setting up a network among the cue and target words (Mera et al., 2021). In the pretesting paradigm and other semantic cue-target pairings, an error could help form a more meaningful relationship between the cue and target. For example, one may generate the word “swims” as a free associate in response to the cue word “whale” when the target word is actually “tail.” Instead of simply encoding the pair “whale” and “tail,” the individual may use the error to create a more robust network between the two words, perhaps thinking of a whale using its tail for swimming. The underlying idea is that prompted retrieval of the target word following the presentation of a cue word evokes several semantically related items. Merging these concepts forms an elaborative memory trace at the time of encoding, which is more likely to be retrieved later at subsequent cue presentations (Huelser & Metcalfe, 2012; Karpicke, 2017). One important finding supporting this theory is that weakly associated word pairs produce stronger retrieval learning than strong associates (Carpenter, 2009). Participants are more likely to retrieve the target word in response to a cue word if they are strongly related. In contrast, weakly related word pairs prompt participants to generate many related words to recall the target. Elaboration enhances future retrieval by assuming individuals encode additional semantically related items alongside the cue and target words.

A primary strength of this theory is its congruence with spreading activation theories of memory. In a spreading activation model, semantic relations link words together in the mental lexicon. The presentation of a cue word activates its corresponding node in the lexical web, with activation spreading to all the nodes connected or semantically related to it. Related words will be activated based on the strength of their association with the cue word (Collins & Loftus, 1975). Thus, the retrieval of a specific memory is not only a function of the frequency and recency of its presentation or base-level activation, but its degree of relation to the current context or spreading activation (Anderson, 1983). Within the framework of this theory, it is clear how elaborative encoding could interact with an already established semantic lexical network. However, debate remains surrounding the actual mechanisms of such encoding and their effects on retrieval.

A standard critique regarding the elaborative hypothesis is its incompatibility with retrieval-induced forgetting, in which retrieval involves the suppression of non-target items, not the activation of several related items (Anderson et al., 1994). Additionally, one can argue that the generation of related items may produce interference or cue overload with subsequent retrieval attempts. However, recent findings suggest that encoding processes may be more dynamic in a way that favors elaborative learning– reactivating older memories while encoding new ones significantly reduces memory interference at retrieval. fMRI data has revealed that this reactivation facilitates the integration of overlapping memories during encoding, thus reducing the interference of otherwise competing memories at retrieval (Chanales et al., 2019). Integration of memories during encoding could reduce interference of memories during retrieval in various contexts, including semantic learning.

The Mediator Hypothesis

The mediator hypothesis proposes that post-error learning occurs because the error acts as a secondary cue to retrieve the correct answer. In a paired-associate task, generating a non-target word related to the cue could mediate between the cue and target words (Huelser & Metcalfe, 2012; Mera et al., 2021). Instead of solely using a cue during retrieval, one can retrieve the error from the cue and the target from the error. Referring to the previous example, at subsequent presentations of the word "whale," one may recall their previous error, "swims," and from it, the correct target word, "tail."

This theory finds its strength in an episodic context account of memory retrieval. The episodic context account explains that people encode information about learning events and the episodic and temporal context in which they occur (Howard & Kahana, 2002). This episodic context may be restored during retrieval to facilitate correct recall (Lehman & Malmberg, 2013). In retrieval-based learning, retrieval increases recall because individuals think back to and then reinstate their prior learning contexts (Karpicke, 2017). Interestingly, retrieval practice in a list discrimination task led free-recall test performance to follow the temporal order of words more closely than the semantic relatedness of words (Whiffen & Karpicke, 2017). Such findings demonstrate that retrieval practice has clear implications for search strategies during future recall, specifically in a manner conducive to the temporal and episodic context of the learning event.

Developing Computational Models of Post-Error Learning

These hypotheses point to distinct mechanisms of post-error learning. Examining these mechanisms could establish one as superior to the other or support their coexistence since, theoretically, they are not mutually exclusive. One way to do this is with the use of formal

computational frameworks (Farrell & Lewandowsky, 2010; Fum et al. 2007). Here, we use Anderson and Schooler's (1991) model, which is now a part of the Adaptive Control of Thought–Rational (ACT-R) cognitive architecture. ACT-R can be seen as a formal implementation of the Multiple Trace Theory of memory (Nadel et al. 2000). In this implementation, long term declarative memories are represented as structured records of information with associated *traces*, where each trace represents a specific episode in which the corresponding information was encoded. The retrievability of each memory *m* depends on its base-level activation B_m , a scalar quantity that reflects the summed contributions of its decaying traces, as indicated in Equation 1:

$$B_m = \ln\left(\sum_{j=1}^{n} t_j^{-d} \right) \tag{1}$$

where n is the number of traces associated with memory m; t_j is the time since the creation of the j-th trace of m; d is a "decay" parameter reflecting how quickly memories are forgotten over time.

In addition to this base-level activation, the probability of retrieving *m* is also a function of spreading activation and noise. Spreading activation reflects the summed contributions of environmental cues that aid retrieval (Collins & Loftus, 1975). Altogether, chunks compete for retrieval based on the relative strengths of base level and spreading activation, as seen in Equation 2:

$$A_m = B_m + \sum_j W_j S_{j,m} + \varepsilon \tag{2}$$

where the sum of j refers to the potential retrieval cues that spread activation to memories, and $S_{j,m}$ is the strength of the association from source j to memory m. Lastly, ε reflects random activation noise that makes the retrieval process non-deterministic; noise is distributed according to a logistic distribution with a mean of zero and a standard deviation π*s / √3, with s being a free parameter ACT-R accurately models forgetting and errors to predict real-world response times and accuracies using these formulas. In this way, ACT-R can comprehensively model a paired-associate task, producing results that closely fit human behavioral data (Anderson, 1981; Anderson & Reder, 1999; Pavlik & Anderson, 2005). Modifying components within ACT-R can create models that test both hypothesized mechanisms of post-error learning.

Spreading activation in ACT-R can be used to properly model elaborative encoding via error commission in a paired-associate task. ACT-R's declarative memory module is formatted in such a way that includes the semantic relationships between memories. Within declarative memory, single words link to various associates in a lexical semantic network. When a memory is retrieved and results in a correct answer, a new trace is added to that memory. But when an error is committed and feedback is provided, a new memory is created that links the cue and target *and* the cue and error words in a single episode. In the previous paired-associate example, the erroneous paired associated *whale-tail* would merge with *whale-swims* to create a composite memory: *whale-tail-whale-swims*. Subsequent presentations of the retrieval cue *whale* spreads more activation to this elaborative memory, because it internally contains two references to the cue word (associated with the correct and incorrect answer) (Figure 1A). Overall, this elaborative encoding of the error alongside the cue and target increases its activation, specifically through spreading activation.

In addition to the declarative module, ACT-R uses a procedural module to articulate discrete cognitive steps (Anderson et al., 2004). The mediator hypothesis relies on remembering the error itself, suggesting that an additional cognitive step occurs when remembering and recalling an error. Thus, a production rule that checks for an error can model a mediator explanation of post-error learning (Figure 1B). If a previous error commission is detected or remembered, an additional production can fire to retrieve the error and use it as a secondary cue.

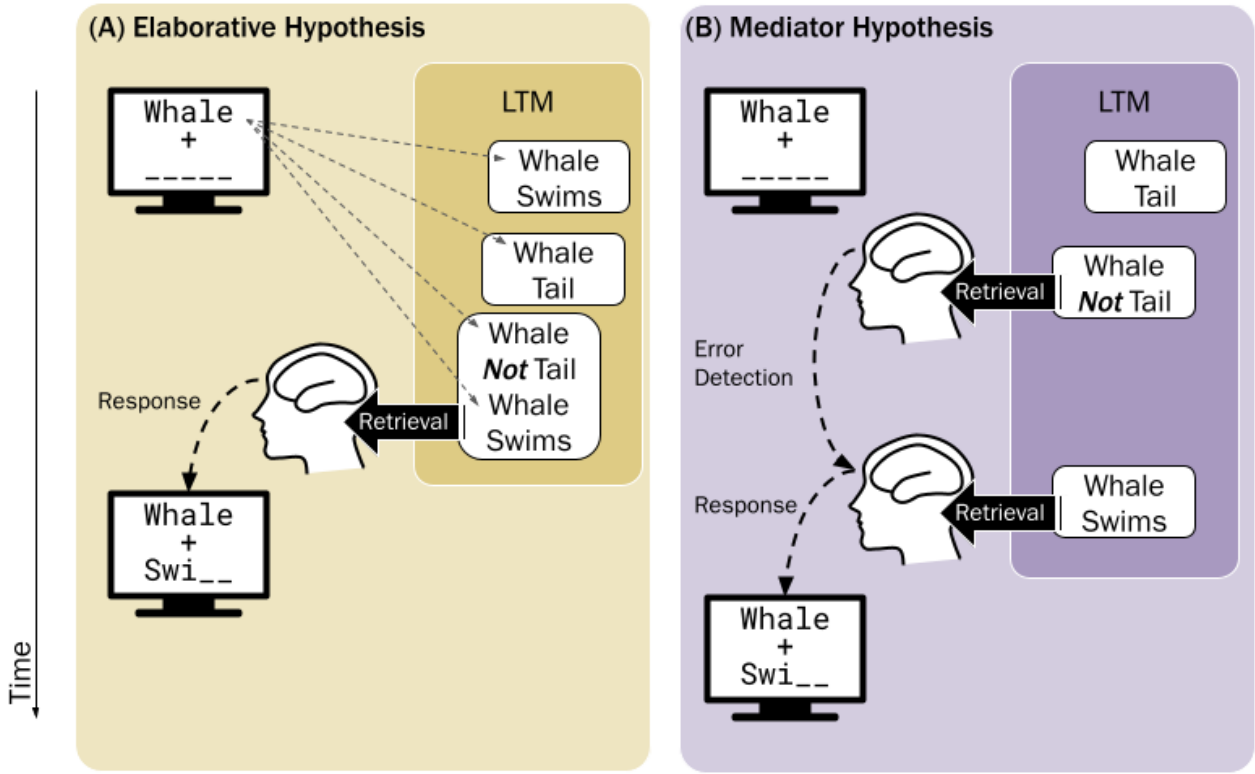

Figure 1: *Theoretical models implementing the two hypothesis for post-error learning*. (A)

Using spreading activation to model elaborative error learning. (B) Using an additional, error-detection cognitive step to model mediator error learning.

Theoretical Predictions Derived from the Models

Even without implementing them, it is possible to derive ordinal predictions from these models. Both models predict that error items would be retrieved better than normal study items. In the case of the elaborative model, this is due to the additional spreading activation and, in the case of the mediator model, to the existence of two retrieval routes. As a result of their increased retrievability, both models predict that error items would be associated with greater response accuracy.

The two models, however, make *opposite* predictions about the relative response times associated with study and error items, respectively. In ACT-R, response times depend on multiple factors, including non-retrieval times spent on perceptual and motor processes, indicated as T_{ER}, and the retrieval time associated with an item i, indicated as $R(i)$. Thus, in general, the response time for the i-th item is:

$$RT = T_{ER} + R(i) \qquad (3)$$

In turn, $R(i)$ depends on the activation $A(i)$ of item i, which is the sum of its base-level and spreading activation. Specifically, retrieval times are related to activation by the equation (Anderson et al., 2004):

$$R(i) = \lambda\, e^{-A(i)} = \lambda / e^{A(i)} \qquad (4)$$

Where λ is another individual-specific parameter that scales the retrieval latency.

Both models assume that all study items have been encoded in the same session and practiced the same number of times, so they have comparable activations. The two models make different predictions for the times to retrieve an error item *e*.

In the *elaborative* model, the additional information encoded in the error item provides additional spreading activation, which sums up the global activation of the error item $A(e)$. We will indicate this additional activation as $S(e)$ so that $A(e) = A(i) + S(e)$. So, the retrieval time for an error item $R(e)$ is:

$$\begin{aligned} R(e) &= \lambda\, e^{-[A(i)+S(e)]} \\ &= \lambda \,/\, e^{[A(i)+S(e)]} \\ &= [\lambda \,/\, e^{A(i)}] \,/\, e^{S(e)} \\ &= R(i) \,/\, e^{S(e)} \end{aligned} \tag{5}$$

Note that, because $S(e) > 0$, $e^{S(e)} > 1$, and thus $R(e) < R(i)$.

According to the *mediator* hypothesis, error items do not differ in terms of activation but in terms of retrieval attempts. That is, on a fraction *f* of trials involving error items, participants would first retrieve an incorrect target, then they would detect the error, and finally retrieve the correct item. Both the correct and incorrect items would have comparable activation levels and thus take approximately the same retrieval time as a study item, $R(i)$. Thus, if we indicate the fraction of trials *f* in which an error is retrieved, we obtain

$$\begin{aligned} R(\mathrm{e}) &= (1 - f)\, R(i) + f\, [R(i) + R(i)] \\ &= (1 + f)\, R(i) \end{aligned} \tag{6}$$

Because $0 < f < 1$, response time will be *longer* for error items, with the specific amount depending on f.

Thus, although both models leave much room for individual differences across participants (due to the T_{ER}, $S(e)$, λ, and f parameters), they make clearly opposite predictions about the relative time to respond to study and error items.

Experimental Predictions

Based on the previous theoretical analysis, we can make the following predictions. Firstly, we expect to confirm the results of previous pretesting research (Huelser & Metcalfe, 2012; Kornell et al., 2009). That is, participants should perform better on error-generation items compared to study items on the final test of our first experiment.

Additionally, we expect to find a difference in response times on the final test between conditions. However, we are unsure about the directionality of this difference. Longer reaction times in the error condition suggest the majority of participants are learning from errors in a mediator method. Shorter reaction times in the error condition suggest the majority of participants are learning from errors in an elaborative method. Finally, by fitting each model to each participant trial-by-trial using maximum log-likelihood, we expect to gain additional insight into error learning mechanisms. Specifically, we may find out whether certain participants are better fit by mediation and others, elaboration (i.e., mediator learners vs elaborative learners).

Methods

Participants

Undergraduate students (N = 61, aged 18-27, 42 female) were recruited on a rolling basis over the course of a quarter for the pretesting task and provided with course credit for their participation. Informed consent was obtained from every participant before the beginning of

every experimental session. Recruitment and experimental procedures were approved by the local Institutional Review Board (Application #00016542).

Pretesting Task

To replicate Huelser and Metcalfe (2012), 60 weakly related word pairs were selected from Nelson, McEvoy, and Schreiber's (1998) norms. Similar to Kornell et al.'s (2009) selection criteria, the forward association of the word pairs was within .05 to .054 with a backward association of 0. This indicates a weak association as 5% of the participants in Nelson and colleagues' (1998) study correctly produced the target word as their free associate when presented with the cue word.

This experiment had three phases: learning, distractor, and final test, and was implemented with the software PsyToolkit (Stoet, 2010, 2017). In the learning phase, the task randomly interleaved study trials and test trials. In study trials, the cue word (e.g., "whale") and its corresponding target ("tail") were presented simultaneously on the screen for 10 seconds. On test trials, only the cue word was presented on the screen (e.g., "whale"). Participants were asked to respond by typing what they thought the target word was in a textbox (e.g., "swims"). They were given 5 seconds to respond before they were shown the cue word and correct target word simultaneously for 5 seconds. After learning all pairs once, participants played a visuospatial game for 5 minutes as a distractor to prevent rehearsal of word pairs. Finally, participants took a self-paced final test containing all 60 word pairs.

Dependent Variables

The primary dependent variable of interest is the accuracy of items on the final test, divided by condition (error or study).

The second dependent variable is the response time to each item on the final test, divided by condition. Response times were calculated as the interval between the onset of each stimulus presentation and the time of the first recorded keystroke in a participant's answer.

Experimental Results

Replicating the Pretesting Effect

On average, participants had higher final cued-recall accuracy on error items (M = 0.70 +/– 0.16) than study items (M = 0.60 +/– 0.18), as seen on the left-hand side of Figure 2, which reports the mean participant-level accuracy across conditions. To account for variability and individual differences, the data was analyzed using a mixed linear model. Specifically, we fitted a mixed model to all of the experimental trials, including the particular trial condition (error or study) as a fixed effect and the participant-level intercept as a random effect; the latter accounts for individual differences in response accuracies. Because accuracy is a binary variable, the model used a binomial linking function to capture the predicted variable. The model uncovered a large main effect of condition ($\beta = -0.47$, $SE = 0.08$, $t = -6.13$, $p < 0.0001$). The complete results of the model are shown in Table 1, and the participant-level predictions are shown in the right-hand side of Figure 2. These findings confirm the results of previous studies (Huelser & Metcalfe, 2012; Kornell et al., 2009).

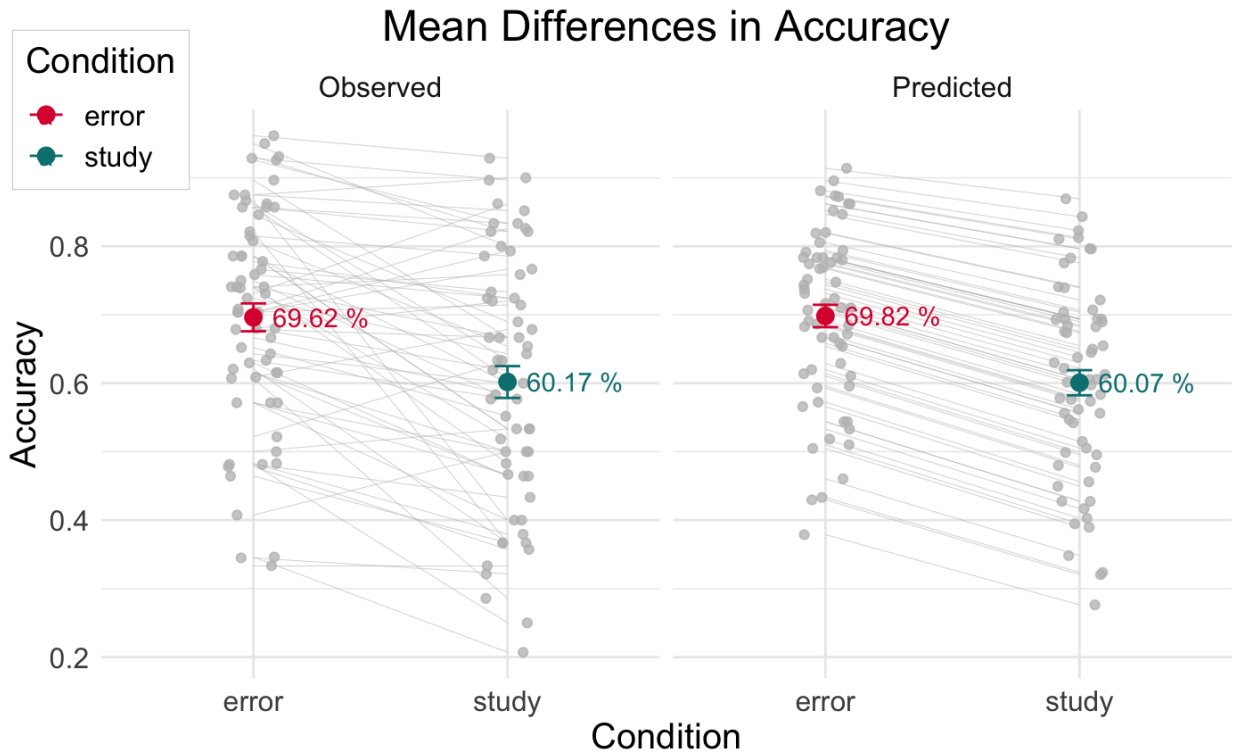

Figure 2 *Differences in average final cued-recall accuracy split by condition.* Gray dots and lines represent data for individual participants; colored dots and error bars represent means +/- SE for the Error (red) and Study (teal) conditions. (Left) mean empirical data; (Right) predicted results from the mixed-linear model.

Table 1

Results of the Mixed-Level Model for Accuracy

Statistical Test	β estimate	*SE*	*t*	*p*
(Intercept)	0.925***	0.107	8.651	5e-18
Condition	- 0.468***	0.076	-6.130	8e-10

$*p < 0.05$ $**p < 0.01$ $***p < 0.001$

Analysis of Response Times

To remove extreme values from our data, we used a maximum cutoff point of 15,000ms and a minimum cutoff point of 200ms. Only correct trials were included in the analysis.

On average, participants had longer response times on error items (M = 4,104 +/– 779ms) than study items (M = 3,920 +/– 936ms), as seen on the left-hand side of Figure 3. The difference between condition response times was compared with a mixed linear model. As in the previous case, the model includes each trial condition (error or study) as a fixed effect and the participant-level intercept as a random effect; the latter accounts for individual differences in baseline response latencies. Unlike the previous case, the dependent variable was modeled using a Gaussian distribution.

Figure 3 *Differences in final cued-recall response times split by condition.* Gray dots and lines represent data for individual participants; colored dots and error bars represent means +/ SE for the Error (red) and Study (teal) conditions. (*Left*) empirical data from the experiment; (*Right*) Predictions of the mixed linear model.

The model confirmed a large and significant main effect of condition ($\beta = -255.99$, $SE = 89.18$, $t = -2.87$, $p < 0.005$). The complete results of the model are shown in Table 2.

To examine the possibility that different individuals might use different strategies, a second mixed linear model was created that included the participant-level slope as a random effect. This model allows for different individuals to have either shorter or longer RTs in the error conditions, thus allowing the possibility that some individuals might use an elaborative

strategy. This second model replicated the results of the first, finding a significant main effect of the condition ($\beta = -255.99$, $SE = 89.18$, $t = -2.87$, $p < 0.005$). An ANOVA test confirmed that the second model does not provide a greater fit than the first ($\chi(3) = 0.38$, $p > 0.94$); furthermore, all the fitted slopes in the ensuing model were negative, suggesting that the apparent upward slopes in Figure 3 are better explained as due to outlier responses, rather than to the systematic use of an elaborative strategy.

Table 2

Results of the Mixed-Level Model for Response Times

Statistical Test	β estimate	*SE*	*t*	*p*
(Intercept)	4109.70***	103.06	39.877	2e-16
Condition	-261.40**	88.79	- 2.944	0.003

*$p < 0.05$ **$p < 0.01$ ***$p < 0.001$

Computational Modeling Results

Although the experimental results are strongly in favor of the mediator hypothesis, considerable variability exists in the data, as shown in the averages of Figures 2 and 3. Despite allowing for the fit of individual data, MLM models are by nature poorly fit to capture the nonlinear dynamics of memory. We are also interested in seeing precisely how well each model fits each individual to examine individual differences comprehensively. For this reason, we implemented the two hypotheses as running ACT-R models and used convex optimization techniques to maximize their fit to every individual.

The fit of the two models was compared by estimating their likelihoods given the data. The likelihood $L(M, \boldsymbol{\theta} \mid \mathbf{x})$, of a model M with parameters $\boldsymbol{\theta}$, given a vector of data $\mathbf{x}$, is the

probability that model M with parameters θ will generate the data θ— that is, $L(M, \boldsymbol{\theta} | \mathbf{x}) = P(\mathbf{x}|M, \boldsymbol{\theta})$. Remember that participants only learned and were tested on each item once to minimize correlation between responses. Thus, our data consists of a vector of multiple independent responses $x_1, x_2, \ldots, x_N$, to different stimuli. Under these conditions, the probability that the model would generate the data $\mathbf{x}$ is the product of the probabilities that the model would generate each individual response:

$$L(M, \boldsymbol{\theta} \mid \mathbf{x}) = P(x_1|M, \boldsymbol{\theta}) \cdot P(x_2|M, \boldsymbol{\theta}) \cdot \cdots \cdot P(x_N|M, \boldsymbol{\theta}) = \prod_i P(x_i|M, \boldsymbol{\theta})$$

Because the product of probabilities becomes vanishingly small, it is common to use *log*-likelihood:

$$\log L(M, \boldsymbol{\theta}|\mathbf{x}) = \log \prod_i P(x_i|M, \boldsymbol{\theta}) = \sum_i \log P(x_i|M, \boldsymbol{\theta})$$

In our case, each model was jointly fitted to two behavioral measures for each stimulus: the response and its corresponding response time. The following sections describe how the response and response time probability densities were computed for the two models. The joint probability distribution of the two variables were created by multiplying their two distributions.

Responses Probabilities

The retrieval probability $P(x)$ of a specific response x can be computed using the Boltzmann equation:

$$P(x) = e^{A(x)/s} \,/\, \sum_i e^{A(i)/s} \qquad (7)$$

Where s is the parameter that controls the standard deviation of the activation noise (Equation 2) , and the sum in the denominator refers to all of the *competing* memories for a given cue, including errors and potentially associated concepts. To get a realistic representation of the

pool of competing memories, all of the responses produced by all participants for a given word were included as potential distractors in the model and modeled as encoded immediately after the beginning of the experiment (at a simulated time of 0.01s), resulting in a minimal activation value for all of them.

In the case of the elaborative hypothesis, Equation 7 can be directly applied to the given response x. In the case of the mediator hypothesis, a correct response can be made either by directly retrieving the correct response or by first retrieving the error-generated response. Thus, the probability of observing a correct response is given by $P(x=correct) + P(x=error)$.

Response Time Probabilities

The probability of observing a given response time for the response x can be computed from Equation 4 if the distribution of the noise term s is known. In ACT-R, declarative memory noise follows a logistic distribution with a mean of zero and a standard deviation of $\pi * s/\sqrt{3}$ (Anderson et al., 2004). Applying Equation 4 to the probability density function of the noise gives the probability density function of different response times, as shown in Figure 4:

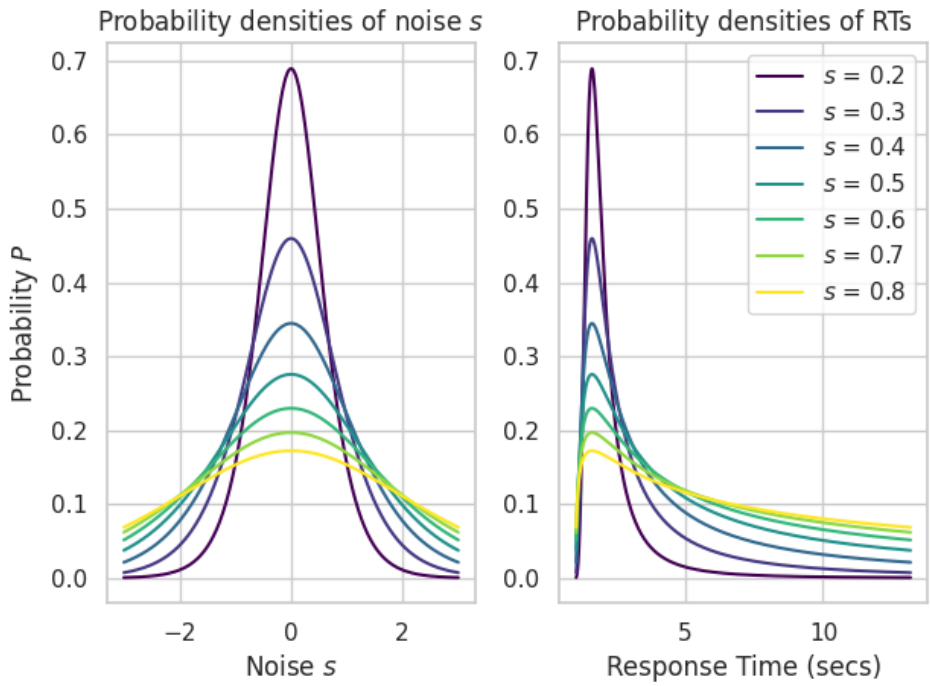

Figure 4*: Different response time probability density functions for different levels of noise s*

Model Fitting

Each model was fit to all the choices of every participant by identifying the combination of parameters values **θ** that maximize the log-likelihood function. Because no closed-form solutions exist to derive the maximum likelihood of ACT-R equations, the best-fitting parameter values were identified using Powell's (1964) optimization algorithm as implemented in Python's SciPy package (Virtanen et al., 2020). This method was chosen because it does not require explicit derivatives and allows us to specify meaningful bounds for parameter values.

Using Powell's method, three parameters were fit for each individual: the decay rate d (which is known to vary significantly across people: Sense et al., 2016), the standard deviation of noise s, and the non-decision time T_{ER}. All of these parameters were constrained within the bounds of [0, 2]. Parameters W (Equation 3) and λ (Equation 4) were kept constant to a default value of $W = \lambda = 1$. This was done because of the parameter identifiability problem: W is difficult to separate from d in Equation 2, and λ has similar effects to T_{ER} in Equation 4.

To perform the trial-by-trial analysis, the models were implemented directly in Python, rather than using the ACT-R architecture. In a way that mimicked the behavioral task, for each participant on each trial (i.e., cue presentation), the model calculated the probability of retrieving the participant's actual response given word activations within the model's declarative module as well as the probability of the participant's response time given word activations and the number of retrievals. Thus, the elaborative model assigned higher probabilities to shorter response times on error items due to spreading activation, while the mediator model assigned higher probabilities to longer response times on error items due to multiple retrievals.

Results

After fitting the values of the d, T_{ER}, and s parameters and computing the maximum log-likelihoods for each participant, we found the greatest majority of participants were best fit by the mediator model ($N = 55$) as seen in Figure 5.

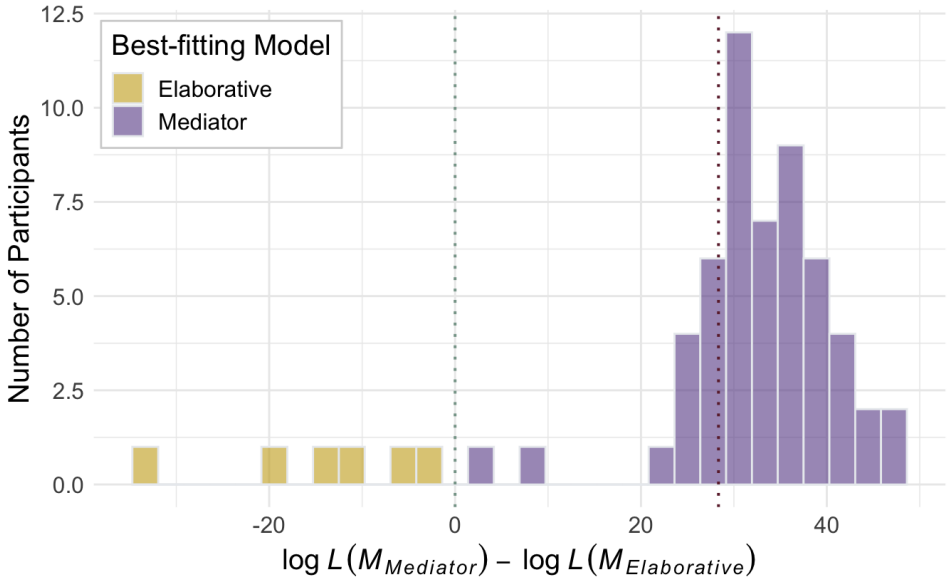

Figure 5. *Histogram log-likelihood differences between the mediator and elaborative models across all participants.* Gold indicates participants who are best fit by the elaborative, and purple

by the mediator, models. The green dotted line indicates the point at which the difference is zero; the red line indicates the mean log-likelihood difference difference (27.9) across all participants.

As the figure shows, the distribution of log-likelihood differences is largely unimodal, with a peak centered to the right of the zero point, corresponding to the area in which the mediator model is preferred over the elaborative model.

For a small number of participant ($N = 6$), the log-likelihood difference was negative, implying that they were best fit by the elaborative model. This finding could reflect either true individual differences (that is, the fact that a small number of individuals do, in fact, behave as predicted by the elaborative hypothesis) or just random noise in the data. These possibilities could be adjudicated by examining the absolute log-likelihoods of the respective model fits: if both models fit the data of these participants abnormally bad, then their differences are more likely to be ascribed to noise. Figure 6 visualizes each participant on a plane where the two dimensions correspond to the absolute log-likelihoods of the elaborative and mediator model. As expected, most participants form a compact cluster above the identity line (grey dotted line in Figure 6), in the are of the plot where the mediator model is more likely than the elaborative model. A small group of eight participats was close to the identity line, suggesting the their behavior did not fall squarely into either of the two models' predictions. Follow-up analysis revealed that, when compared to the others, these participants did not necessarily conform to the predictions of the elaborative model. Instead, compared to the other participants, they exhibited lower accuracies (51.8% vs. 61.1%) and higher response times (4,707ms vs. 3,834ms) on *study* items, while their performances on the error items was undistinguishable from that of the rest of rest of participants. We concluded that the inability of the mediator model to best fit these

partticipants was not due to the superiority of the elaborative model but to a greater proportion of errors and abnomally slow responses that was not easily fit by either model.

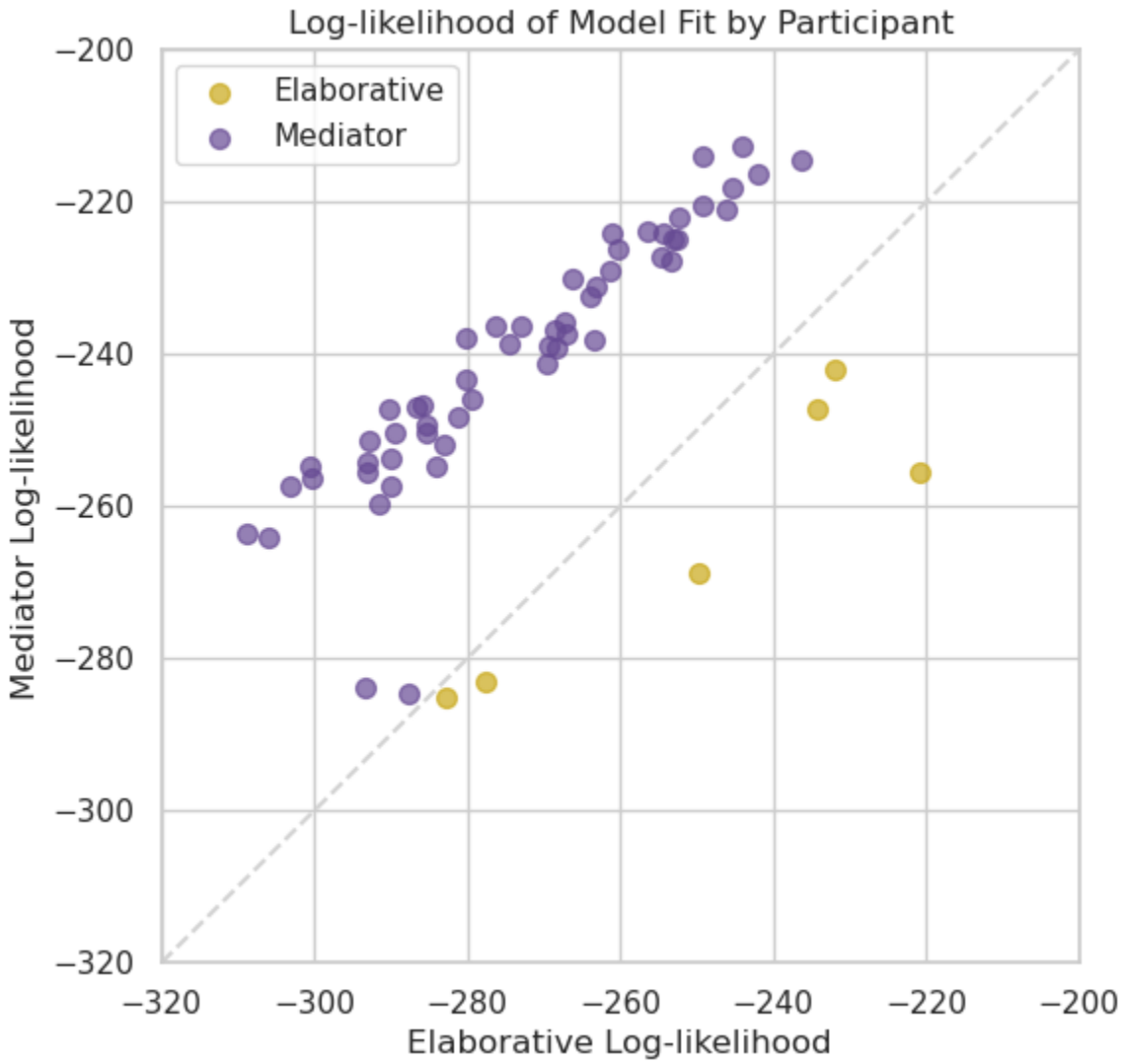

Figure 6. *Scatterplot of individual participants and the absolute log-likelihood values for the Elaborative (x-axis) and Mediator (y-axis) models.* Gold points represent participants who are best fit by the elaborative, and purple by the mediator, models. The grey dotted line is the identity line, where both models have the same log-likelihood.

The group-level evidence for two models can be computed by aggregating their participant-level differences (Yang & Stocco, 2021). In this case, the cumulative difference in log-likelihood between the two models is 1,728. Because of the definition of log-likelihood, this difference is equivalent to an odds ratio of $e^{1,728}$, indicating that the mediator model is $e^{1,728}$ more likely to fit the sample data than the elaborative model.

Discussion

In this paper, we examined whether errors facilitate learning, including an investigation to probe potential underlying mechanisms. The pretesting paradigm reflects the benefit of a retrieval process where participants guess an answer in response to a cue before studying the correct cue-answer pair. This paradigm is commonly used in error learning research. As such, its success in improving learning compared to just studying is well documented (Huelser & Metcalfe, 2012; Kornell et al., 2009). However, research into the underlying cognitive processes facilitating this phenomenon remains speculative. Without understanding how post-error learning works, applications remain restrictive. Revealing underlying mechanisms may allow us to discriminate when errors are beneficial. In this way, current learning and memory models could increase their efficacy by harnessing the power of errors. Our study aims to provide a baseline to research post-error learning mechanisms with the use of cognitive models.

Importantly, we were able to replicate the pretesting results of existing literature; final cued-recall accuracy was higher in the error-generation condition than in the study-only condition. This not only helps to confirm the benefit of retrieval attempts before study opportunities but advocates for further research into the mechanisms of this process.

Different reaction time hypotheses arose from our two post-error learning models based on the ACT-R architecture. First, an elaborative model predicts that error learning results in quicker response times on subsequent tests. This is because elaboration works through spreading activation, adding activation to the correct answer which speeds up retrieval and response times. Alternatively, a mediator model predicts that error learning results in slower response times on subsequent tests. Mediation uses an extra step to retrieve an error as a secondary cue to get the correct answer. Although this procedure increases accuracy, it also costs extra time, resulting in longer response times. Results from the current study demonstrate that average response times are longer on error items than on study items. This supports the mediator hypothesis of post-error learning. However, it is important to note that at the individual participant level, there are cases in which average reaction times on study items are longer than on error items. This could mean that error learning mechanisms differ on an individual level. Future research should attempt to distinguish between mechanisms by focusing on individual differences. Additionally, the ability to learn from errors itself may be highly individualized. Focusing on individual differences to look at (1) who is learning from errors, and (2) how they are doing it can better specify post-error learning and its applications to learning environments.

The most notable limitation of this study is the generalizability of the error-learning paradigm. Errors as we think of them in the real world are often committed after a study opportunity. Looking not only at retrieval errors, but encoding errors, in error learning would broaden the applications to more settings. Overall, it's important to extend error learning research to paradigms that involve real-world errors. Another limitation of this paradigm is it does not look at performance over longer periods of time. Testing memory for error trials over days, weeks, and months, may reveal stronger benefits of errors. For example, it is possible that

the mediator process best explains performance for the first test after an error, but the elaborative model would provide a better explanation for subsequent repetitions of the stimulus. It is also possible that deeper encoding processes resulting from error commission may lead to facts that are more resistant to decay and forgetting. To address these possibilities, future research would need to adopt alternative paradigms designed and used to further establish post-error learning beyond retrieval practice. An example of such paradigm is the adaptive fact-learning system developed by Sense and van Rijn (2022; Sense et al., 2016; Zhou et al., 2021), in which new paired associations are presented at a pace that is individualized to each participant to optimize retention. Importantly, their paradigm internally makes use of ACT to model each individual's memory, and yields highly reliable estimates of each individual and each item's decay rate (Sense et al., 2016). A modification of this paradigm that includes an error-generating phase provides important information as to whether, for example, error items are forgotten at lower speeds, rather than (or in addition to) having additional retrieval routes.

Models are unique in their ability to reconceptualize behavioral results. By decoding human behavior, models begin to reveal cognition by stabilizing the messiness of data. As such, the proposed cognitive models in this paper can help identify mechanisms of post-error learning. These models could distinguish different learners from one another and propose ways to manipulate post-error learning by targeting the relevant cognitive processes. Moreover, these findings could extend to fields outside of cognitive psychology, advocating for the benefit of making mistakes in various educational settings and assisting in developing AI and machine learning advancements that update comprehensive feedback histories with each new learning experience.

Declaration of Interest

Declarations of interest: none

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